

Geophysics III

Calculus Tutorial Solutions

March 9, 2016

1 Derivative Practice Problems

Problem 1.1. Plate Cooling

To find the heat flow from the temperature, we need to take the derivative of the temperature with respect to depth; the solution includes the following derivative

$$\frac{d}{dz} \frac{1}{n} \sin\left(\frac{n\pi z}{L}\right)$$

Solution 1.1.

$$\frac{d}{dz} \frac{1}{n} \sin\left(\frac{n\pi z}{L}\right)$$

$$\begin{aligned} \frac{d}{dx} au &= a \frac{d}{dx} u \\ a &= \frac{1}{n} \\ u &= \sin\left(\frac{n\pi z}{L}\right) \end{aligned}$$

$$\frac{1}{n} \frac{d}{dz} \sin\left(\frac{n\pi z}{L}\right)$$

Problem 1.2. Seismic Reflection

To find the minimum travel-time due to a seismic reflection off a boundary, we need to take the derivative of $t(x)$ as below,

$$\frac{d}{dx} v_1^{-1} [(h^2 + x^2)^{1/2} + (h^2 + (d - x)^2)^{1/2}]$$

Problem 1.3. Decompression Melting

To model partial melting, you need to take the derivative of the melt depression term,

$$\frac{d}{dF} K \left(\frac{X_0}{D + F(1 - D)} \right)^\gamma.$$

Problem 1.4. Gravity on an Ellipsoid

As part of the derivation to determine the gravity of an ellipsoid (remember we make an ellipsoidal correction to gravity), we need to differentiate the equation of an ellipsoid,

$$\frac{d}{d\theta} \frac{1 - f}{1 - f \sin^2 \theta}. \quad (1)$$

2 Integral Practice Problems

Problem 2.1. Steady-State Temperature

To find the temperature in a half-space, we need to integrate the thermal gradient with depth,

$$\int \left(\frac{q_0}{k} - \frac{A}{k} z \right) dz,$$

where q_0 , A , and k are constants.

Problem 2.2. Radioactive Decay

To find the number of parent radionuclides present due to radioactive decay, we need to integrate both sides of the following equation,

$$\int_{N_0}^N \frac{dN'}{N'} = - \int_0^t \lambda dt',$$

and once complete, solve for N .

Problem 2.3. Buried Slab

This integral is similar to one we needed for our buried slab problem,

$$\int_0^R \frac{r}{(r^2 + z^2)^{3/2}} dr.$$

Problem 2.4. Mass Flux

The mass flux of asthenosphere (mass passing a given point per unit time) moved by a tectonic plate can be expressed as

$$\int_0^h \rho u_0 \frac{y}{h} dy.$$

Problem 2.5. Entropy

In petrology, the entropy is a key thermodynamic parameter. The change in entropy is related to temperature as

$$\Delta S = \int_{T_1}^{T_2} \frac{C_P}{T} dT. \quad (2)$$